Bonald, Ziyue LI

W2 Professor in Machine Learning in Smart Cities
Department of Operations & Technology, Technical University of Munich, Germany

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Google Scholar: https://scholar.google.com/citations?user=q5_My2AAAAAJ&hl=en

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Research Interests

My research interests are: **spatiotemporal machine learning for complex AI systems, especially smart transportation**. I'm ranked as **Top-50 Researcher** in "spatiotemporal data mining" and "smart mobility" by Google Scholar.

Our works have been published as 30+ papers in top-tier AI conferences (NeurIPS, KDD, IJCAI, ICDE, CIKM, AAAI, ICLR), 10 papers in top journals (TITS, TKDE, TKDD, JMLR), awarded with **9 best paper awards**, and also deployed in real industries as **well-proven products** in **smart mobility**.

Methodology:

- Spatiotemporal statistics: high-dimension data, tensor decomposition, graphical models;
- Deep learning: LLMs, self-supervised learning, transfer/meta-learning, reinforcement-learning;

Major Applications

- smart mobility and smart city: prediction, clustering, anomaly detection, causal learning, traffic control systems, traffic congestion management
- large-scale spatial-temporal systems: supply chain, logistics, time-series, energy, earth system

Key Scientific Metrics

- Publications: 41 (peer-reviewed) + 17 (under review)
- Citations: 1000+ • h-index: 20 • i10-index: 30 (Google Scholar) • Paper Awards: 9
- Third-party Funding: 345,665 / 698,560 USD (since UoC)
- Scholarship and Individual Award: 175,247 USD (at HKUST and UoC)

Academic Experience

W2 Professor @ Technical University of Munich — 08.2025 - present
Department of Operations & Technology, Heilbronn Data Science Center — *Heilbronn, Germany*
W1 Professor @ University of Cologne — *03.2022 - present*
Department of Information Systems (Top 1 in Germany by AIS) — *Cologne, Germany*

- ***Research****: Conduct research in spatiotemporal data mining for smart and sustainable mobility and cities; with 6 refereed journal papers, 23 referred conference papers, and 16 working papers during UoC; Won three Best Paper Awards during UoC; Supervising one visiting scholar, co-supervising three Ph.D.s, and supervised 10 master/bachelor thesis;*
- ***Teaching****: Designed four new courses from scratch; highly evaluated from 4.2/5.0 to 5.0/5.0.*

Chief Machine Learning Scientist @ EWI (Energy Economics Institute) — *03.2022-12.2023*
University of Cologne — *Cologne, Germany*

- *Research outputs: IEEE PES ISGT'24 [C21], Nature Energy'24 [R3]*
- *Teach outputs: Co-supervise three Ph.D.s, and two master thesis.*

Research Visit & Collaboration @ Georgia Institute of Technology — *01.2023*
School of Industrial and Systems Engineering, Georgia Institute of Technology — *Atlanta, GA, U.S.A*

- *Research: NeurIPS'24 [R14], IEEE TKDE'24 [R6]*
- *Teaching: Lecture of "Advanced Machine Learning for Smart Mobility" (Host: Prof. Jianjun Shi)*

Education Background

The Hong Kong University of Science and Technology[1] *Hong Kong*

[1] About HKUST: QS Ranking: #27, Data Science: #10, Engineering: #18 worldwide

Ph.D. in Industrial Engineering and Decision Analytics *09.2017 - 08.2021*

- Thesis: "Incorporating Domain Knowledge into Big Data: with Application in Smart Manufacturing and Transportation" (2nd Runners-Up in Best Thesis Competition), and Advisor: Prof Fugee Tsung

- Hong Kong PhD Fellowship Award and Excellent Research Award (Highly Prestigious)

Arizona State University *Phoenix, AZ, U.S.A*

Joint PhD Supervision in School of Computing and Augmented Intelligence *02.2019 - 08.2021*

- Research: Tensor Decomposition and Graphical Models, and Advisor: Prof. Hao Yan

- Awards: Six best paper awards in INFORMS data mining and statistics.

Xi'an Jiaotong University[2] *Xi'an, China*

[2] About XJTU: QS Ranking: #251, Mechanical Engineering: #1 in China

Bachelor of Engineering in Mechanical Engineering (Major in Data Analytics) *06.2013 - 06.2017*

Bachelor of Economics in Finance *06.2015 - 06.2017*

- Outstanding Graduates Award, National Scholarship Award

University of New South Wales *Sydney, Australia*

Bachelor Research Visiting in Mechanical and Manufacturing Engineering *2015*

- Concentration: Data Analytics-based Fault Detection, and Advisor: Dr Wade Smith

- GPA: HD (High Distinction), highest possible score

International Awards and Prizes

1. ***Best Paper Award*** *for the paper "Enabling Tensor Decomposition for Time-Series Classification via A Simple Pseudo-Laplacian Contrast", Institute of Industrial and Systems Engineers (IISE), Quality Control and Reliability Engineering (QCRE) (2025).*
2. ***Peter Luh Young Researcher Award (Runner-up)*** *for the paper "Dynamic Causal Graph Convolutional Network for Traffic Prediction", selected 3/600, highly prestigious, IEEE Robotics and Automation Society (2023).*
3. ***Best Paper Award (Runner-up)*** *for the paper "Graph-aware Tensor Topic Models for Individualized Passenger Travel Pattern Clustering", IISE, QCRE (2023).*
4. ***Best Student Paper Award*** *(Finalist) for the paper "Tensor Completion for High-dimensional Data with Graph-structured and Weakly-dependent Sample Dimension", IISE QCRE (2022).*
5. ***Best Applied Paper Award (Winner)*** *for the paper "Tensor Topic Models with Graphs and Applications on Individualized Travel Patterns" INFORMS Data Mining and Decision Analytics (DMDA) Workshop (2021).*
6. ***Best Theoretical Paper Award (Runner-Up)*** *for the paper "Low-Rank Robust Subspace Tensor Clustering for Metro Passenger Flow Modeling", INFORMS DMDA (2021).*
7. ***Best Conference Paper Award (Winner)*** *for the paper "Long-Short Term Spatiotemporal Tensor Prediction for Passenger Flow Profile", IEEE International Conference on Automation Science and Engineering (CASE) (2020): selected from 1/500, highly prestigious.*
8. ***Best Student Poster Award*** *(Finalist) for the paper "Profile Decomposition based Hybrid Transfer Learning for Cold-start Data Anomaly Detection", INFORMS Quality, Statistics, and Reliability (QSR) Section (2020).*
9. ***Best Student Paper Award*** *(Finalist) for the paper "Individualized Passenger Travel Pattern Multi-Clustering based on Graph Regularized Tensor Latent Dirichlet Allocation", INFORMS Data Mining (2020).*
10. ***Best Student Paper Award*** *(Finalist) for the paper "Tensor Completion for Weakly-Dependent Data on Graph for Metro Passenger Flow Prediction", INFORMS QSR Section (2020).*

Research Grant and Proposal Writing Experience

Research Grants:

1. "Towards Efficient Electricity Market based on Meta Reinforcement Learning" ***(Under Review)***
   - Agency/Company: DFG
   - Role: Co-PI, together with Prof. Wolfgang Ketter (University of Cologne, Germany)
   - Amount: 500,000 Euro (535,442 USD), Candidate Share: 50%
   - Intended Period of Contract: 01.06.2025 - 30.05.2028
   - Details: This project focuses on developing a meta-reinforcement-learning-based agent for smart grid and energy trading under various market mechanisms and dynamics. My team will focus on the algorithm design of the meta agent, and Prof. Ketter's team will focus on market simulation and mechanism design. I led the team in project ideation, task allocation, and proposal writing.
2. "Scalable Climate Disaster Damage Assessment by Graph Foundation Model using Remote Sensing Imagery" ***(Under Review)***
   - Agency/Company: Amazon Research Award
   - Role: Sole PI
   - Amount: 100,000 USD
   - Intended Period of Contract: 01.07.2025 - 31.06.2026
   - Details: This project aims to develop a scalable climate risk assessment system based on pre- and post-disaster remote sensing images. My team will focus on the whole project. I solely contribute to project ideation, task allocation, proposal writing, and will lead project supervision.
3. "Towards A Robust and Generalizable Mobility Data Analytical Foundation based on Noisy 3D Geospatial Perception" ***(Under Last-stage Review, 1st Review with High-ranked Rating)***
   - Agency/Company: Sony Research Award Program
   - Role: Sole PI
   - Amount: 100,000 USD
   - Intended Period of Contract: 01.06.2025 - 31.05.2026
   - Details: This project focuses on developing a robust and generalizable trajectory model based on "perception-action-interation" contrastive learning and meta-learning, for navigating in a 3D geospatial enviroment. My team will focus on the whole project. I solely contribute to project ideation, task allocation, proposal writing, and project supervision and management.
4. "On Developing Video Traffic Data based Intelligent Transport Fundamental Core Technique" ***(On-going)***, ***ID: NGLM-011***
   - Agency/Company: S-Lab[1]
   - Role: Co-PI, together with Prof. Cheng Long (Nanyang Technological University, Singapore)
   - Amount: 867,000 SGD (656,388 USD), Candidate's Share: 50%
   - Period of Contract: 15.03.2023 - 14.06.2025
   - Details: This project focuses on a deep learning-based solution for improving vision-based traffic data quality, including mission data completion, kriging, and prediction. Both My team and Prof. Long's team focus on ideation, model/algorithm design, and experiments. Our team particularly also focuses on industrial feasibility, deployment, and project management. Both I and Prof Long contributed to the proposal writing and presentation. The output includes publications in IEEE MDM'24 [C23], IEEE TKDE'24 [J8], NeurIPS'24 [R9, R13, R16], ECCV'24 [C18].
5. "Intelligent Traffic Congestion Decision Systems based on Spatiotemporal and Multi-modal Data" ***(Finished), ID: STTHU/2204***
   - Agency/Company: S-Lab
   - Role: Co-PI, together with Prof. Chen Zhang (Tsinghua University, China)
   - Amount: 300,000 CNY (42,172 USD), Candidate's Share: 50%
   - Period of Contract: 01.05.2022 - 01.05.2023
   - Details: This project focuses on causal inference models to better explain and analyze traffic congestion patterns. Both I and Prof. Zhang's team focus on ideation, proposal composition,

[1] S-Lab: Research Funding Agency that promotes research collaboration between Asian Artificial Intelligence Universities (e.g., in Singapore, China, Hong Kong), industry, and international institutions outside of Asia

model/algorithm design, experiments, and project management. Our team particularly particularly deployed the research outputs as industrial products. The research output includes publications in IEEE CASE'23 [C7], KDD'23 [C6], JASA'24 [R4].

Proposal Writing at the Hong Kong University of Science and Technology

1. ***Foshan-Hong Kong Technology Funding***, *"Multi-stage process monitoring and optimization based on big data analytics and machine learning", 2021-2023*
   *- **PI**: Prof. Fugee Tsung (Ziyue Li, Yinghui Huang); **Grant**: 4,811,634 HK$ (619,198 USD), **ID**: FSPM02202003.*
2. ***Innovation and Technology Commission***, *"Trial: A Big Data Trial Scheme for Smart Public Transportation Crowd Monitoring ", 2019-2021*
   *- **PI**: Prof. Fugee Tsung (Ziyue Li, Man Li, Zhenli Song); **Grant**: 1,175,300 HK$ (151,246 USD), **ID**: ITT/007/19GP.*
3. ***RGC - General Research Fund***, *"Statistical Transfer Learning with Applications to Quality Control and Monitoring", 2019-2021*
   *- **PI**: Prof. Fugee Tsung (Ziyue Li, Ke Zhang, Kai Wang, Longwei Chen, Zhenli Song); **Grant**: 632,421 HK$ (81,384 USD), **ID**: RGC-16201718.*
4. ***RGC - General Research Fund***, *"Statistical Learning, Prediction, and Monitoring Methods for Urban Rail Transit Systems", 2018-2020*
   *- **PI**: Prof. Fugee Tsung (Zhenli Song, Ziyue Li, Kai Wang, Yinghui Huang); **Grant**: 443,950 HK$ (57,130 USD), **ID**: RGC-16203917.*

Scholarships and Individual Funding

***Since Ph.D.**:*

1. ***ACM Young Researcher Travel Grant 2023 (Winner)*** ***Amount**: 1,000 USD*
   - Funding Agency: ACM SIGSPATIAL (2023).
2. ***HK PhD Fellowship Scholarship** (2017-2020)* ***Amount**: 1,296,000 HKD (166,686 USD)*
   - Highly selective and prestigious, with only 2 recipients in the department.
   - Funding Agency: Hong Kong Research Grants Council
3. ***HKUST Excellent Research Award** (2017)* ***Amount**: 40,000 HKD (5,144 USD)*
   - Funding Agency: HKUST
4. ***Best Thesis Award 2021 (2^{nd} Runners-up)*** ***Amount**: 3,000 HKD (356 USD)*
   - HKUST Three Minute Thesis Competition.
5. ***1^{st} Runners-up, Audience Award**, Hackathon@UST 2018* ***Amount**: 15,000 HKD (1,929 $)*

***During Bachelor Study**:*

1. ***National First Prize**, "Challenge Cup" National Curricular Academic Science and Technology Contest: A data-driven study for land transfer in rural China 2015*
2. ***Honored Graduate Award**, Xi'an Jiaotong University 2017*
3. ***Regional First Prize**, 7th Mechanical Innovation & Design Competition 2016*
4. ***National Scholarship**, Xi'an Jiaotong University 2016*
5. ***1st-Runners-up Award**, XJTU Entrepreneurship Competition 2015*
6. ***National Encourage Scholarships**, Xi'an Jiaotong University 2014, 2015*

Ongoing Research Projects

"Tensor Decomposition for Metro System's Demand and Transit Data" *08.2017 - present*
Collaborating with Arizona State University, HKUST, Tsinghua University, etc

- *Tensor models with semantic graphs for spatiotemporal demand prediction, completion, and anomaly detection, with publications in ACM SIGSPATIAL'24 [C11], IEEE CASE'23 [C9], IEEE CASE'20 [C2], AAAI'20 [C1], INFORMS Journal on Data Science'24 [J9], and IEEE RA-L'20 [J3].*
- *Tensor topic models with semantic graphs for spatiotemporal behavior clustering, with publications in ACM SIGSPATIAL'24 [C12], ICDE'21 [C3], Data Mining and Knowledge Discovery'22 [J4].*

- *Tensor Self-Supervised Learning: with paper in NeurIPS'24 [R13], IEEE Transactions on Knowledge and Data Engineering'24 [R4] under review.*

"Robust and Explainable Deep Learning for Transport Systems" *10.2021 - present*
Collaborating with TU-Wien, Baidu Research, GaTech, HKUST, Stanford, etc

- *Self-supervised learning for more generalizable and robust spatiotemporal analysis, with publications in AISTATS'24 [C18], IEEE CASE'23 [C10, C8], ACM MobiSys'23 [C5], CIKM'22 [C4], ACM Transactions on Knowledge Discovery from Data'22 [J5].*
- *Spatiotemporal causal structure learning to explain traffic systems, with publications in IEEE CASE'23 [C7], SIGKDD'23 [C6] and JASA'24 [J10].*

"Generalized AI Agents for Traffic Signal Control and Decision Automation" *06.2022 - present*
Collaborating with Baidu Research, Nanyang Technological University

- *Generalist and foundational agents based on Meta-RL for cross-scenario decision-making, specifically traffic signal control, with publications in SIGKDD'24 [C22], IJCAI'24 [C21], AAMAS'24 [C20, C19], IEEE TKDE'24 [J8], IEEE TITS'24 [J7]', and IEEE TITS'24 [R2] under review*

"Large Language Models for Spatiotemporal Time-Series" *08.2023 - present*
Collaborating with Nanyang Technological University, HKUST, ST Research

- *Developing LLMs for time-series, including prediction, classification, time-series + other modalities (e.g., text), with publications in IEEE MDM'24 [C25], and NeurIPS'24 [R15, R14] and ACM Computing Surveys'24 [R7] under review.*

"Large Language Models for Industrial Interdisciplinary Application" *08.2023 - present*
Collaborating with Roche, Baidu Research, ST Research

- *AutoTask: LLMs for complex task planning and tool usage, with publications in ICLR'24 [C15, C14]*
- *AutoCode: LLMs for coding (Text-to-SQL, Text-to-Python), with publications in IEEE DEXA'24 [C16], and NeurIPS'24 [R10, R9] under review.*

Key Research Initiative "Sustainable Smart Energy and Mobility" *03.2022 - present*
Collaborating with Institute of Energy Economics of UoC

- *data-driven methods for personalized and smarter mobility: with publications in ACM SIGSPATIAL'23 [C12, C11], KDD'23 [C6], IEEE T-ITS'24 [J7]; and NeurIPS'24 [R12, R11] under review.*
- *Towards explainable energy management under external tension (e.g., Ukraine-Russia War): with publications in IEEE PES ISGT Europe'24 [C24]; and NeurIPS'24 [R6] under review.*

Industrial Experience

My three-year industry experience gives me the unique advantage of sharp sense for industrial needs, delivering high-quality and applicable research, multi-project management ability, and leadership.

Data Mining Researcher *08.2021 - 03.2022*
Smart Mobility Group @ Hong Kong Science Park *Hong Kong*

- *Led a team of 10 researchers (3 full-time, 5 interns, 1 project manager), conducted front-tier research in spatiotemporal analysis, and delivered 5+ pipelines of AI-driven products, including:*
  - *One LLM-based Text-to-SQL system for large-scale transportation database query*
  - *One LLM-based agent system to automate complex corporate task planning and tool usage*
  - *One machine learning-based traffic data infrastructure platform*
  - *Traffic Signal Control Systems based on optimization-based and Reinforcement Learning: deployed in an actual city, with 26% higher efficiency.*
  - *One causal inference-based Traffic congestion management and cause analysis.*
  - *Developed our own LLMs for traffic domain knowledge and traffic data analysis.*
- *Corresponding research work published in IJCAI, CIKM, KDD, AAAI, ICLR, AAMAS, TMM, and more, with topics including self-supervised learning, reinforcement Learning, and resource-efficient domain adaptation, transfer learning, and meta-learning.*

Research Project Manager *09.2020 - 08.2021*
Hong Kong Metro MTR Co.-HKUST R&D Project *Hong Kong*

- *Research collaboration with MTR and Transport Department (HK), developing an Intelligent Transport System (ITS) based on data-driven methods, with publications in AAAI, ICDE, SIGSPATIAL, and DAMI.*
- *Delivered the solution for demand prediction, station clustering, and passenger pattern mining.*

Cloud Computing Scientist *09.2019 - 02.2020*
Nokia Bell Labs *Stuttgart, Germany*

- *Research in serverless computing, machine learning systems based on Amazon Web Service (AWS), and Bell Labs KNIX MicroFunctions.*
- *Conducted data transmission latency analysis, serverless system component profiling and optimization, and serverless ML inference and training (NLP, CV models) in AWS and KNIX.*

Internal University Committee Service

- *Search Committee of W1 Professorship for Smart Sustainable Energy, University of Cologne, 2024*
- *Committee Member of Study Strategy for Generative AI, University of Cologne, 2024*
- *PostDoc Recruitment Committee, Cologne Institute of Information System, 2024*
- *Graduate Admission Committee for Ph.D., Key Research Initiatives (KRI) of Sustainable Smart Mobility, Summer 2024*
- *Marketing Committee, Cologne Institute of Information Systems, Winter 23/24 - present*

International Professional Services

Council Member:

- *Council Member of INFORMS Quality Statistics Reliability (QSR) Section 2025, Nominated*
- *Sub-Council (Public Communication) Member of INFORMS QSR Section 2024*
- *Sub-Council (Public Communication) Member of INFORMS Data Mining Section, 2024*

Conference / Workshop Organizer:

- *Anomaly Detection with Foundation Models, with Bosch AI and Mitsubishi, @ IJCAI 2024*
- *The Second Workshop on Vision-based Industrial Inspection, with Apple, @ ECCV 2024*
- *CVPR Workshop on Vision-based Industrial Inspection, with Apple, @ CVPR 2023*
- *Spatial and Temporal Modeling for Decision Intelligence, with ASU, @ IEEE CASE 2023*

Challenge Organizer:

- *Data Challenges on ECCV Workshop on Vision-based Industrial Inspection, @ ECCV 2024*
- *Data Challenges on CVPR Workshop on Vision-based Industrial Inspection, @ CVPR 2023*

Conference Session Chair:

- *IJCAI 2024: Session "Urban Computing"*
- *KDD 2023: Session "Spatiotemporal Machine Learning"*
- *INFORMS Annual Meeting 2023: Sessions "Structure Learning from Heterogeneous Data"*
- *INFORMS Annual Meeting 2023: Sessions "Spatiotemporal Decision Intelligence"*
- *IEEE CASE 2023: Sessions "Spatiotemporal Decision Intelligence"*
- *CIKM 2022: Session "Temporal Data"*
- *CIKM 2022: Session "Clustering"*
- *INFORMS Annual Meeting 2021: "Knowledge-integrated Data-driven Smart Transportation"*

Professional Membership:

- *ACM SIGSPATIAL*
- *IEEE Intelligent Transportation Systems Society*
- *IEEE Power and Energy Society*
- *INFORMS Data Mining Section*
- *INFORMS Quality, Statistics, Reliability Section*
- *Institute of Industrial and Systems Engineers (IISE) Quality Control & Reliability Engineering (QCRE) Division*

- *Transportation Statistics Group of ASA*

Editorship:

- *Guest Editor, IISE Transactions, Special Issue on "Advances in Reinforcement Learning for Intelligent Systems"*

Conference Reviewing Service:

- *IJCAI 2025, AAAI 2021-2025, AAMAS 2025, NeurIPS 2024, IJCAI 2024, SIGKDD 2023, CVPR 2023, ECAI 2023, ECML-PKDD 2022-2023, ICRA 2022-2024, ICIS 2023, IEEE CASE 2020-2023, INFORMS Best Paper Competition 2023*

Journal Reviewing Service:

- *INFORMS Journal of Data Science*
- *Transportation Research Part C (TR-C)*
- *GIScience Remote Sensing*
- *IEEE Transactions on Intelligent Transportation Systems (T-ITS)*
- *IEEE Transactions on Automation Science and Engineering (T-ASE)*
- *IEEE Internet of Things Journal*
- *IET Software*
- *Springer Pattern Analysis and Applications (PAAA)*
- *IISE Transactions*
- *Technometrics*

International Cooperations

*I demonstrate a strong and international collaboration personality, where I often take the leadership role in collaboration projects. * indicates a strong connection.*

Cooperations within Germany and Europe

- *Prof. Wolfgang Ketter*, University of Cologne, Germany*
- *Prof. Youness Dehbi*, HafenCity University Hamburg, Germany*
- *Dr. Hao Li, TU Munich, Germany*
- *Prof. Oliver Ruhnau, University of Cologne, Germany*
- *Prof. Stefano Canali, Politecnico di Milano, Italy*
- *Prof. Alessandra Angelucci, Politecnico di Milano, Italy*

Cooperations with the U.S. and Canada

- *Prof. Michael Lepech, Stanford University, CA*
- *Prof. Jan Jianjun Shi*, Georgia Institute of Technology, Atlanta, GA*
- *Dr. Man Li*, Georgia Institute of Technology, Atlanta, GA*
- *Prof. Lijun Sun*, McGill University, Montreal, Canada*
- *Prof. Hua Wei*, Arizona State University, Phoenix, AZ*
- *Prof. Hao Yan*, Arizona State University, Phoenix, AZ*

Cooperations with Top Universities in Asia

- *Prof. Cheng Long*, Nanyang Technological University (NTU), Singapore*
- *Prof. Fugee Tsung*, The Hong Kong University of Science and Technology, Hong Kong*
- *Prof. Chen Zhang*, Tsinghua University, China*

Cooperations with Industries

- *Dr. Ruichuan Chen, Distinguished Member of Technical Staff, The Bell Labs*
- *Dr. Rui Zhao*, Chief Research Executive, SenseTime Research*
- *Dr. Meng Cao, Tech Lead Manager, Apple*
- *Dr. Mohammadreza Fani Sani*, Data Scientist, Microsoft*

Selected Invited Talks

1. *"Spatiotemporal Machine Learning for Complex Database System"* *10.2024*
   Ruhr University Bochum *Germany*
2. *"From Perception to Decision: Robust and Generalizable Spatiotemporal Learning"* *10.2024*
   The Hong Kong University of Science and Technology *Hong Kong*
3. *"Meta Reinforcement Learning-based Agent for Cross-City Traffic Signal Control"* *08.2024*
   The 1st AI for Critical Infrastructure Workshop @ IJCAI-24 *Jeju, South Korea*
4. *"Multi-city Co-trained Agent for More Robust Traffic Signal Control"* *08.2024*
   The 3rd International Workshop on Spatio-Temporal Reasoning and Learning *Jeju, South Korea*
5. *"Robust Tensor Decomposition with Self-Supervised Learning for Noisy Time-Series"* *08.2024*
   The 1st Workshop Anomaly Detection with Foundation Models *Jeju, South Korea*
6. *"Large Language Model in Time-series Analysis"* *05.2024*
   The 2nd INFORMS Conference on Quality, Statistics, and Reliability *Milan, Italy*
7. *"Large Language Model for Smart Energy"* *03.2024*
   Key Research Initiative of Sustainable Smart Energy & Mobility Workshop *Cologne, Germany*
8. *"Spatiotemporal Self-Supervised Learning for Time-Series Analysis"* *10.2023*
   INFORMS Annual Meeting *Phoenix, Arizona*
9. *"Towards a City-Agnostic Traffic Signal Control Agent"* *10.2023*
   INFORMS Annual Meeting *Phoenix, Arizona*
10. *"Advanced Statistics and Deep Learning in Smart Mobility"* *08.2023*
    Tsinghua University *Beijing, China*
11. *"Advanced Deep Learning in Smart Mobility"* *02.2023*
    Georgia Institute of Technology *Atlanta, U.S.A*
12. *"High-dimensional Statistics in Smart Mobility"* *01.2023*
    Georgia Institute of Technology *Atlanta, U.S.A*
13. *"Spatiotemporal Self-Supervised Learning for Time-Series Prediction"* *05.2023*
    Institute of Industrial and Systems Engineers Annual Meeting *New Orleans, U.S.A*
14. *"Graph Tensor Topic Models for Individual Travel Analysis"* *05.2023*
    Institute of Industrial and Systems Engineers Annual Meeting *New Orleans, U.S.A*
15. *"Spatiotemporal Self-Supervised Learning for Road Network and Trajectory"* *10.2022*
    3rd Workshop on Data-driven Intelligent Transportation @ CIKM-22 *Atlanta, U.S.A*
16. *"Tensor Topic Models for Individual Travel Analysis"* *10.2022*
    INFORMS Quality, Statistics, Reliability Flash Paper *Indianapolis, U.S.A*
17. *"Tensor Topic Models with Graph and Applications in for Individual Travel Analysis"* *4.2021*
    ICDE PhD Symposium *Chania, Greece (Online)*
18. *"Tensor Latent Dirichlet Allocation Models with Graph for Individual Travel Clustering"* *10.2020*
    INFORMS Annual Meeting *Maryland National Harbor, U.S.A*
19. *"Statistical Transfer Learning for Cold-Start Time-Series Anomaly Detection"* *10.2020*
    INFORMS Annual Meeting *Maryland National Harbor, U.S.A*
20. *"Graph Tensor Decomposition for Spatiotemporal Traffic Flow Prediction"* *02.2020*
    Data Science Symposium in Waseda University *Tokyo, Japan*
21. *"Graph Tensor Decomposition for Spatiotemporal Traffic Flow Prediction"* *10.2019*
    INFORMS Annual Meeting *Seattle, U.S.A*
22. *"Statistical Multi-Task Learning for Cold-Start Time-Series Anomaly Detection"* *10.2018*
    INFORMS Annual Meeting *Phoenix, U.S.A*

List of Publications

Summary

- ***Overall***: ***41*** *peer-reviewed publications, of which* ***28*** *as corresponding author,* ***7*** *as first author*
- ***High-Quality***: ***16*** *are in A*-ranked top-tier AI conferences/journals (e.g., IJCAI, AAAI, ICLR, NeurIPS, SIGKDD);* ***7*** *are in A-ranked top-tier AI conferences/journals (e.g., CIKM, AISTATS).*
- ***Impact***: *since past 4 years, the Google Scholar Citation:* ***1000+****, h-Index:* ***20****, i10-Index:* ***30***
- ***Conference***: ***31*** *peer-reviewed conference proceedings*
- ***Journal***: ***11*** *papers in journals with peer review, of which 11 international; 1 paper in journal reviewed by the editorial board*
- ***Working Paper***: ***17*** *papers are under review.*

- Notes: The ∗ sign indicates corresponding author; underlined authors are my mentees; I am highlighted in **boldface**. *Listed in chronological order. Blue color is clickable link to the paper.*

Refereed Conference Publications

Year: 2020

[C1] **Z Li***, N D Sergin, H Yan, C Zhang, and F Tsung*, "Tensor Completion for Weakly-Dependent Data on Graph for Metro Passenger Flow Prediction" AAAI Conference on Artificial Intelligence (**AAAI 2020**). *Oral, CORE Rank A* in computer science (A* - 7.47% of 803 ranked venues, very prestigious).*

- ***Best Student Paper Award*** (Finalist), INFORMS Quality, Statistics, and Reliability, 2020

[C2] **Z Li***, H Yan, C Zhang and F Tsung*, "Long-Short Term Spatiotemporal Tensor Prediction for Passenger Flow Profile" in IEEE 16th International Conference on Automation Science and Engineering (**IEEE CASE 2020**).

- ***IEEE CASE 2020 Best Conference Paper Award***, IEEE Robotics & Automation Society.

Year: 2021

[C3] **Z Li***, "Tensor Topic Models with Graphs and Applications on Individualized Travel Patterns", 2021 IEEE 37th International Conference on Data Engineering (**ICDE 2021**). *Oral, CORE Rank A*.*

- ***Best Applied Paper Award (Winner)***, INFORMS Data Mining and Decision Analytics, 2021

Year: 2022

[C4] Z. Mao, **Z. Li***, D. Li, L. Bai, & R. Zhao. "Jointly Contrastive Representation Learning on Road Network and Trajectory". The 31st ACM International Conference on Information & Knowledge Management (**CIKM 2022**). *CORE Rank A in Computer Science (A - 14.4% of 803 ranked venues)*

Year: 2023

[C5] H Ruan, Q Gong, Y Chen*, J Chen, **Z Li**, X Su. "A Contrastive-Prototypical Privacy-preserving Heart Rate Prediction System for Drivers in Connected Vehicles". The 21st Annual International Conference on Mobile Systems, Applications and Services (**MobiSys 2023**). *CORE Rank A.*

[C6] T. Lan, **Z. Li***, Z. Li, L. Bai, M. Li, F. Tsung, W. Ketter, R. Zhao, and C. Zhang. "MM-DAG: Multi-task DAG Learning for Multi-modal Data - with Application for Traffic Congestion Analysis". In Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**SIGKDD 2023**). *Oral, CORE Rank A*.*

[C7] J. Lin, **Z. Li***, Z. Li, L. Bai, R. Zhao and C. Zhang, "Dynamic Causal Graph Convolutional Network for Traffic Prediction", 2023 IEEE 19th International Conference on Automation Science and Engineering (**IEEE CASE 2023**), Auckland, New Zealand.

- ***Peter Luh Young Researcher Award***, IEEE Robotics and Automation Society, 2023.

*[C8] L Wang, L Bai, **Z Li***, R Zhao and F Tsung, "Correlated Time Series Self-Supervised Representation Learning via Spatiotemporal Bootstrapping", IEEE 19th International Conference on Automation Science and Engineering (**IEEE CASE 2023**), Auckland, New Zealand.*

*[C9] M. Jiang, A. Wang, **Z. Li***, and F. Tsung, "A Unified Probabilistic Framework for Spatiotemporal Passenger Crowdedness Inference within Urban Rail Transit Network", 2023 IEEE 19th International Conference on Automation Science and Engineering (**IEEE CASE 2023**), Auckland, New Zealand.*

*[C10] Y. Chen, **Z. Li**, W. Ouyang and M. Lepech*, "Adaptive Hierarchical SpatioTemporal Network for Traffic Forecasting", 2023 IEEE 19th International Conference on Automation Science and Engineering (**IEEE CASE 2023**), Auckland, New Zealand.*

*[C11] D Li, **Z Li***, Z Li, L Bai, Q Gong, L Sun, W Ketter, and R Zhao. "A Critical Perceptual Pre-trained Model for Complex Trajectory Recovery". International Conference on Advances in Geographic Information Systems (**ACM SIGSPATIAL 2023**). CORE Rank A.*

- ***ACM SIGSPATIAL Young Researcher Travel Grant Award, Winner.***

*[C12] **Z Li***, H Yan, C Zhang, L Sun, W Ketter, and F Tsung. "Tensor Dirichlet Process Multinomial Mixture Model with Graphs for Passenger Trajectory Clustering". International Conference on Advances in Geographic Information Systems (**ACM SIGSPATIAL 2023**). CORE Rank A;*

- ***Best Paper Award (Runner-up) in the QCRE division of IISE 2023***

[C13] J Ruan, Y Chen, B Zhang, Z Xu, T Bao, G Du, S Shi, H Mao, **Z Li**, X Zeng, and R Zhao. "TPTU: Large Language Model-based AI Agents for Task Planning and Tool Usage". The 37th Conference on Neural Information Processing Systems (**NeurIPS 2023**) - Workshop on Foundation Models for Decision Making. CORE Rank A*.*

Year: 2024

*[C14] Z Li, Y Nie, **Z Li***, L Bai, Y Lv, R Zhao, "Non-Neighbors Also Matter to Spatiotemporal Kriging: A New Contrastive-Prototypical Learning", The 27th International Conference on Artificial Intelligence and Statistics (**AISTATS 2024**). Oral, CORE Rank A.*

[C15] H. Mao, R. Zhao, **Z. Li**, and et al., "PDiT: Interleaving Perception and Decision-making Transformers for Deep Reinforcement Learning", The 23rd International Conference on Autonomous Agents and Multiagent Systems (**AAMAS 2024**). Oral, CORE Rank A*.*

*[C16] J. Lu, J. Ruan, H. Jiang, **Z. Li***, H. Mao and R. Zhao, "DuaLight: Enhancing Traffic Signal Control by Leveraging Scenario-Specific and Scenario-Shared Knowledge", The 23rd International Conference on Autonomous Agents and Multiagent Systems (**AAMAS 2024**). Oral, CORE Rank A*.*

*[C17] H. Jiang, **Z. Li***, X. Xiong, J. Ruan, J. Lu, H. Mao, R. Zhao, "X-Light: Cross-City Traffic Signal Control Using Transformer on Transformer as Meta Multi-Agent Reinforcement Learner", The 33rd International Joint Conference on Artificial Intelligence (**IJCAI 2024**), Oral, CORE Rank A*.*

*[C18] J. Ruan, **Z. Li***, H. Jiang, J. Lu, H. Mao, R. Zhao, "CoSLight: Co-optimizing Collaborator Selection and Decision-making to Enhance Multi-intersection Traffic Signal Control", Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**SIGKDD 2024**), Oral, CORE Rank A*.*

[C19] B Zhang, H Mao, J Ruan, Y Wen, Y Li, S Zhang, Z Xu, D Li, **Z Li***, R Zhao, "Controlling Large Language Model-based Agents for Large-Scale Decision-Making: An Actor-Critic Approach", The Twelfth International Conference on Learning Representations Workshop on LLM Agents (**ICLR 2024**). CORE Rank A*.*

[C20] Y. Kong, J. Ruan, Y. Chen, B. Zhang, T. Bao, S. Shi, G. Du, X. Hu, H. Mao, **Z. Li***, X. Zeng, R. Zhao, "Boosting Task Planning and Tool Usage of Large Language Model-based Agents in Real-world Systems", The 2024 Conference on Empirical Methods in Natural Language Processing (**EMNLP 2024**). CORE Rank A*.*

*[C21] S Yang, Q Su, Z Li, **Z Li***, H Mao, C Liu, R Zhao, "SQL-to-Schema Enhances Schema Linking in Text-to-SQL", The 35th Database and Expert Systems Applications Conferences and Workshops*

(**IEEE DEXA 2024**).

[C22] Q. Xu, G. Lin, C. Loy, C. Long*, **Z. Li**, R. Zhao, "Eliminating Feature Ambiguity for Few-Shot Segmentation", The 18th European Conference on Computer Vision, **ECCV 2024**, CORE Rank A*.

[C23] Q. Xu, X. Liu, L. Zhu, G. Lin, C. Long*, **Z. Li**, Z. Rui, "Hybrid Mamba for Few-Shot Segmentation", The 38th Annual Conference on Neural Information Processing Systems (**NeurIPS 2024**), CORE Rank A*.

[C24] G. Su, S. Yang, Z. Li, **Z. Li***, "Transformer-based Drum-level Prediction in a Boiler Energy Plant with Delayed Relations among Multivariates", IEEE Power & Energy Society (PES) Innovative Smart Grid Technologies (ISGT) Europe Conference 2024.

[C25] P. Kai Peter, Y. Li, **Z. Li***, W. Ketter, "Deep Causal Learning to Explain and Quantify The Geo-Tension's Impact on Natural Gas Market", IEEE Power & Energy Society (PES) Innovative Smart Grid Technologies (ISGT) Europe Conference 2024 (**Supervised Master Thesis**).

[C26] C. Liu, S. Yang, Q. Xu, **Z. Li***, C. Long, Z. Li, R. Zhao, "Spatial-Temporal Large Language Model for Traffic Prediction", The 25th IEEE International Conference on Mobile Data Management (**IEEE MDM 2024**). CORE Rank B.

Year: 2025

[C27] C. Liu, Q. Xu, H. Miao, S. Yang, L. Zhang, C. Long, **Z. Li***, R. Zhao, "TimeCMA: Towards LLM-Empowered Time Series Forecasting via Cross-Modality Alignment", The 39th Annual AAAI Conference on Artificial Intelligence (**AAAI 2025**), CORE Rank A*, accepted.

[C28] Q. Xu, C. Long, **Z. Li***, S. Ruan, Z. Li, R. Zhao, "KITS: Inductive Spatio-Temporal Kriging with Increment Training Strategy", The 39th Annual AAAI Conference on Artificial Intelligence (**AAAI 2025**), CORE Rank A*, accepted.

[C29] Z Li, X Wang, J Zhao, S Yang, G Du, X Hu, B Zhang, Y Ye, **Z Li***, R Zhao, "PET-SQL: A Prompt-enhanced Two-stage Text-to-SQL Framework with Cross-consistency", The 30th International Conference on Database Systems for Advanced Applications (DASFAA 2025), CORE Rank B, accepted.

- Our method is **ranked as the TOP-1 in Text-to-SQL worldwide Leaderboard**.

[C30] M Keltek, R Hu, FS Mohammadreza, **Z Li***, "LSAST-Enhancing Cybersecurity through LLM-supported Static Application Security Testing, The 40th IFIP TC-11 SEC 2025 International Information Security and Privacy Conference, 2025, accepted **(Supervised Bachelor Thesis)**.

[C31] Q. Xu, C. Long, **Z. Li***, S. Ruan, Z. Li, R. Zhao, "Efficient Multivariate Time Series Forecasting via Calibrated Language Models with Privileged Knowledge Distillation", The 2025 IEEE International Conference on Data Engineering (**ICDE 2025**), CORE Rank A*, accepted.

Refereed Journal Publications

Year: 2020

[J1] F Tsung*, **Z Li**, "Discussion of 'A novel approach to analysis of spatial and functional data over complex domains'" Quality Engineering, 2020, published.

[J2] **Z Li***, H Yan, C Zhang, F Tsung, "Long-Short Term Spatiotemporal Tensor Prediction for Passenger Flow Profile" IEEE Robotics and Automation Letters (**IEEE RA-L 2020**). CORE Rank A*.

- **Winner** of IEEE CASE 2020 **Best Conference Paper Award**.

Year: 2022

[J3] A. Angelucci*, **Z. Li,***, N. Stoimenova, S. Canali (2022)." The paradox of artificial intelligence system development process: the use case of corporate wellness programs using smart wearables". AI & Society.

[J4] **Z Li***, H Yan, C Zhang, F Tsung, "Individualized Passenger Travel Pattern Multi-Clustering based on Graph Regularized Tensor Latent Dirichlet Allocation". Data Mining and Knowledge Discovery

*(**DAMI 2022**), Springer. CORE Rank A.*

- ***Best Student Paper Award**, Data Mining, INFORMS 2020, **Finalist Award**.*

[J5] **Z. Li***, H. Yan, K. Zhang and F. Tsung, "*Profile Decomposition based Hybrid Transfer Learning for Cold-start Data Anomaly Detection*". *The ACM Transactions on Knowledge Discovery from Data (**ACM TKDD 2022**). CORE Rank A.*

- ***Best Student Poster Award**, QSR, INFORMS 2020, **Finalist Award**.*

Year: 2023

[J6] K. Liu, S. Tang, **Z. Li***, *et al.*, "*Relation-Aware Distribution Representation Network for Multi-Modalities Clustering*", *in IEEE Transactions on Multimedia (**IEEE MM 2023**). CORE Rank A*.*

Year: 2024

[J7] H Jiang, **Z Li***, Z Li, L Bai, H Mao, W Ketter, R Zhao, "*A General Scenario-agnostic Reinforcement Learning for Traffic Signal Control*", IEEE Transactions on Intelligent Transportation Systems (**IEEE TITS 2024**). *CORE Rank A*.*

[J8] X Du, **Z Li***, *Long C, Xing Y, Philip Yu, Chen H*, "*FELight: Fairness-Aware Traffic Signal Control via Sample-Efficient Reinforcement Learning*", IEEE Transactions on Knowledge and Data Engineering (**IEEE TKDE 2024**). *CORE Rank A*.*

[J9] D Sergin Nurretin, J Hu, **Z Li**, C Zhang, F Tsung, H Yan*. "*Low-Rank Robust Subspace Tensor Clustering for Metro Passenger Flow Modeling*", INFORMS Journal on Data Science (**IJDS 2024**).

- ***Best Theoretical Paper Award (Runner-Up)**, INFORMS Data Mining and Decision Analytics 2021*

[J10] T Lan, **Z Li***, J Lin, Z Li, L Bai, M Li, F Tsung, C Zhang*, "*MultiFun-DAG: Multivariate Functional Directed Acyclic Graph*", Journal of Machine Learning Research, **JMLR 2024**, conditional accept. *CORE Rank A*.*

Book Chapter

[B1] Z. Li. "*Incorporating Domain Knowledge into Big Data: with Application in Smart Manufacturing and Transportation*". *Hong Kong University of Science and Technology (Hong Kong), 2021.*

[B2] Y. Hao, **Z. Li**, X. Zhao, and J. Hu. "*Sparse Decomposition Methods for Spatio-Temporal Anomaly Detection.*" *In* **Multimodal and Tensor Data Analytics for Industrial Systems Improvement**, *pp. 185-206. Cham: Springer International Publishing, 2024.*

Papers under Review

Submitted to Journals:

[R1] Z. Li, **Z. Li***, X. Hu, G. Du, Y. Nie, F. Zhu, L. Bai, R. Zhao, "*VisionTraj: A Noise-Robust Trajectory Recovery Framework based on Large-scale Camera Network*", **IEEE Transactions on Intelligent Transportation Systems (IEEE T-ITS)**, *2024+, under review.*

[R2] H. Jiang, X. Xiong, **Z. Li***, H. Mao, G. Sui, J. Ruan, Y. Cheng, R. Zhao, "*GuideLight: 'Industrial Solutions' Guidance for More Practical Traffic Signal Control Agents*", **IEEE Transactions on Intelligent Transportation Systems (IEEE T-ITS)** *2024+, under review.*

[R3] J. Hu, **Z. Li**, C. Zhang, F. Tsung, H. Yan, *"Tensor Completion for High-dimensional Data with Graph-structured and Weakly-dependent Multi-Sample Dimensions"*, **IEEE Transactions on Automation Science and Engineering (IEEE TASE)** *2024+, under review.*

[R4] M. Li, **Z. Li***, *"Robust Self-Supervised Deep Tensor Decomposition for Corrupted Time Series Data"*, **IEEE Transactions on Knowledge and Data Engineering (IEEE T-KDE)** *2024+, CORE Rank A*, under review.*

*[R5] P. Guo, P. Jin, Y. Zhang, L. Bai, **Z. Li***, "Online Test-Time Adaptation of Spatial-Temporal Forecasting", **IEEE Transactions on Intelligent Transportation Systems (IEEE T-ITS)**, 2024+, under review.*

*[R6] Y. Li, **Z. Li***, O. Ruhnau, P. Peter Kai, W. Ketter, "Predict the New Normal of Natural Gas Market in Germany after the Russia-Ukraine War", **Nature Energy**, 2024+, under review.*

*[R7] J. Ye, W. Zhang, K. Yi, Y. Yu, **Z. Li**, J. Li, F. Tsung "A Survey of Time Series Foundation Models: Generalizing Time Series Representation with Large Language Model", **ACM Computing Surveys** 2024+, under review.*

Submitted to Conferences:

*[R8] L Zhang, L Shen, Y Zheng, S Piao, **Z Li**, F Tsung, LeMoLE: "LLM-Enhanced Mixture of Linear Experts for Time Series Forecasting", The 41st IEEE International Conference on Data Engineering **ICDE 2025**, CORE Rank A*, 2025+, under review.*

*[R9] Y Li, L Wang, S Piao, BH Yang, **Z Li**, W Zeng, F Tsung, "MCCoder: Streamlining Motion Control with LLM-Assisted Code Generation and Rigorous Verification", The 2025 IEEE International Conference on Robotics & Automation **ICRA 2025**, CORE Rank A*, 2025+, under review.*

*[R11] G. Sui, Z. Li, **Z. Li***, S. Yang, J. Ruan, H. Mao, and R. Zhao. "Reboost Large Language Model-based Text-to-SQL, Text-to-Python, and Text-to-Function - with Real Applications in Traffic Domain", **ICDE 2024**, CORE Rank A*, under review.*

*[R12] B Zhang, Y Ye, G Du, X Hu, Z Li, S Yang, CH Liu, R Zhao, **Z. Li***, H Mao, "Benchmarking the Text-to-SQL Capability of Large Language Models", The 38th Annual Conference on Neural Information Processing Systems (**NeurIPS 2024**), CORE Rank A*, under review.*

[R14] Q. Xu, X. Liu, L. Zhu, G. Lin, C. Long, **Z. Li**, Z. Rui, "Hybrid Mamba for Few-Shot Segmentation", The 38th Annual Conference on Neural Information Processing Systems (**NeurIPS 2024**), CORE Rank A*, under review.*

*[R15] X. Guo, **Z. Li***, T. Lan, Z. Li, B. Zhao, C. Zhang, D. Wang, X. Zhang, "XTraffic: A Dataset Where Traffic Meets Incidents with Explainability and More", The 38th Annual Conference on Neural Information Processing Systems (**NeurIPS 2024**), CORE Rank A*, under review.*

*[R16] M. Li, **Z. Li***, L. Sun, F. Tsung, "Enabling Tensor Decomposition for Time-Series Classification via A Simple Pseudo-Laplacian Contrast", The 38th Annual Conference on Neural Information Processing Systems (**NeurIPS 2024**), CORE Rank A*, under review.*

*[R17] W. Zhang, J. Ye, **Z. Li***, J. Li, F. Tsung, "DualTime: A Dual-Adapter Multimodal Language Model for Time Series Representation", The 38th Annual Conference on Neural Information Processing Systems (**NeurIPS 2024**), CORE Rank A*, under review.*

Teaching Experience

Summary:

- ***Teaching at University of Cologne***: *I have two courses per semester on average, each of them is 2 weekly hours. I created three new courses from scratch, i.e., "Seminar Machine Learning", "Advanced Data Analytics", and "Decision Making under Uncertainty".*
- ***Diverse Teaching Portfolio***: *I have taught diverse course for students from diverse background, such as deep learning (course titled "Advanced Data Analytics") to business analytics students, reinforcement learning (course titled "Decision Making under Uncertainty") to information system student, statistics (course titled "Quality Engineering") to management students.*
- ***High Rating***: *Some of my courses, Advanced Seminar Machine learning, have been rated full star 5.0/5.0 or 4.9/5.0 for two consecutive years.*
- ***Teaching Focus***: *My main teaching focus is general machine learning, deep learning, statistics, convex optimization, and so on.*

Lecturing:

At the University of Cologne:

Semester	***Course***	***Level***	***# Enroll***	***Eval.***
Winter 24/25	*Seminar Machine Learning*	*Undergraduate*	*18*	*On-going*
Winter 24/25	*Decision Making under Uncertainty*	*Graduate*	*12*	*On-going*
Winter 24/25	*Advanced Data Analytics*	*Graduate*	*31*	*On-going*
Summer 2024	*Seminar Machine Learning*	*Undergraduate*	*4*	*N/A*
Winter 23/24	*Seminar Machine Learning*	*Undergraduate*	*16*	***4.9/5.0***
Winter 23/24	*Decision Making under Uncertainty*	*Graduate*	*4*	*4.5*/5.0*
Winter 23/24	*Advanced Data Analytics*	*Graduate*	*23*	***4.1/5.0***
Summer 2023	*Advanced Seminar Machine Learning*	*Graduate*	*14*	***5.0/5.0***
Winter 22/23	*Advanced Seminar Machine Learning*	*Graduate*	*3*	*4.9*/5.0*
Winter 22/23	*Decision Making under Uncertainty*	*Graduate*	*3*	*4.2*/5.0*
Summer 2022	*Advanced Seminar Machine Learning*	*Graduate*	*8*	***4.0/5.0***

* *only courses with more than 5 students have official evaluation reports.*

At Georgia Institute of Technology:

- *Guest Instructor,* ***"Advanced Machine Learning for Smart Mobility"*** *(PhD), Winter 2023*
  - *Designed the contents and assignments in "Tensor-based Smart Transportation" and "Deep Learning-based Smart Mobility, Energy, and Earth"*

At EWI gGmbH:

- *Instructor,* ***"Machine Learning Bootcamp"*** *(PhD), EWI gGmbH* — *Winter 23/24*
  - *Took care of the one-week boot camp in machine learning, with the topic in machine learning basics, time-series prediction, anomaly detection, LLM for time-series data, and so on.*

At the Hong Kong University of Science and Technology

- *Teaching Assistant,* ***"Engineering Management"*** *(Undergraduate)* — *Spring, 2020*
- *Co-instructor,* ***"Quality Engineering"*** *(Undergraduate)* — *Spring, 2019*
- *Teaching Assistant,* ***"Six Sigma Quality Management"*** *(Graduate)* — *Fall, 2018*

Student Advising:

- Notes: There are two Ph.D. students and one PostDoc in my lab. My other students mainly come from collaboration and industry, whom I supervised as sincerely and committedly as my own PhD students, with one group discussion and one individual discussion weekly.

Current PhD Students and PostDoc:

- *Rong Hu (Ph.D., candidate)*
  - Topics: Tensor Decomposition and High-dimensional Statistics
  - Supervision Period: 07.2024 - present
- *Philipp Kai Peter (Ph.D., candidate)*
  - Topics: Spatiotemporal Causal Learning
  - Supervision Period: 03.2023 - present
  - Research Output: IEEE PES ISGT Europe Conference [C21], Nature Energy [R4]
- *Man Li (PostDoc; Now: Faculty at Southwestern University of Finance and Economics)*
  - Topics: Tensor Self-Supervised Learning for Time-Series Analysis
  - Supervision Period: 09.2022 - 08.2024
  - Research Output under My Supervision: KDD'23 [C5], TMLR [R6], IEEE TKDE'24 [R11], NeurIPS'24 [R7]

Co-supervised Students from Academic Collaboration:

- *Sun Yang (Msc, Peking University, China)*
  - Topics: LLM-based Decision Intelligence
  - Supervision Period: 10.2023 - present
  - Research Output under My Supervision: IEEE MDM'24 [C17], NeurIPS'24 [R11], ICLR'24 [C16], ICDE'24 [R10]
- *Qianxiong Xu (Ph.D., Nanyang Technological University, Singapore)*
  - Topics: Few-shot Segmentation, Spatiotemporal Learning
  - Supervision Period: 02.2023 - present
  - Research Output under My Supervision: IEEE MDM'24 [C17], KDD'24 [R1], ECCV'24 [R5]
- *Chenxi Liu (PostDoc, Nanyang Technological University, Singapore)*
  - Topics: LLM-based Spatiotemporal Learning
  - Supervision Period: 02.2023 - present
  - Research Output under My Supervision: IEEE MDM'24 [C17]

Supervised Advisees from Industry *(I served as their line manager):*

- *Haoyuan Jiang (Msc, Now: Baidu)*
  - Topics: Reinforcement-Learning-based Traffic Signal Control
  - Supervision Period: 06.2022 - 10.2023 (under my supervision, conducting research from zero to one, and one to excellence!)
  - Research Output under My Supervision: IJCAI'24 [C12], AAMAS'24 [C11], IEEE TITS'24 [J4], KDD'24 [R5], IEEE IROS'24 [R6]
- *Zhishuai Li (Ph.D., S.T. Research; Now: Faculty at The China University of Petroleum)*
  - Topics: Spatiotemporal Learning, Intelligent Transport Systems
  - Supervision Period: 03.2022 - 09.2024
  - Research Output under My Supervision: ICLR'24 [C16], AISTATS'24 [C9], ICDE'24 [R10], SIGSPATIAL'24 [R2], KDD'23 [C5], IEEE CASE'23 [C4], SIGSPATIAL'23 [C3]

Supervised Master and Bachelor Thesis:

- *Julius Junker (Msc in Business Analytics, On-going), "Identifying Optimal Location for Next EV Charging Station based on Multi-Modal Data"*
  - ***Collaboration with German Space Agency (DLR)****.*

- *Mete Keltek (BS in Information Systems, On-going), "Large Language Model (LLM)-based Vulnerability Scanning for Website"*
  *- **The work is under review of 3rd IEEE Conference on Secure and Trustworthy Machine Learning, and will be submitted to the AAAI 2025 Undergraduate Consortium**.*
- *Yulin Li (Msc in Information Systems, Summer 2023), "Deep Causal Learning to Explain and Quantify The Geo-Tension's Impact on Natural Gas Market"*
  *- **This work is accepted in 2024 IEEE PES Innovative Smart Grid Technologies Conference Europe, and the extended work is under review of Nature Energy**.*
- *René Oblonczek (Msc in Business Analytics, On-going), "Enhancing Business Analytics with Cloud-Native Solutions and Large Language Models"*
- *Vu Duc Anh Nguyen (Msc in Business Analytics, On-going), "Leveraging Large Language Models to Predict Energy Price Equilibrium after Major Disruptive Events"*
- *Fabian Guenkel (Msc in Business Analytics, On-going), "Retrieval Augmented Generation - Structuring a landscape conservation plan for time scheduling"*
- *Benedict Bruhn (Msc in Business Analytics, On-going), "Time Series Forecasting with LLMs - Leveraging Multimodal Data for Improved Predictive Modeling"*
- *Servando Pizarro (Msc in Business Analytics, On-going), "Retrieval-Augmented Generation (RAG)-based Annual Report Engineering"*
- *Minh Bui Cong (Msc in Information Systems, Summer 2024), "Spatiotemporal Causal Analysis of Traffic Incident's Impact on Traffic Flows"*
- *Ricarda Beck (Msc in Information Systems, Summer 2024), "Spatiotemporal Prediction for Multi-source International Cargo Demand"*
- *Nils Hornstein (BS in Information Systems, Winter 22/23), "Deep Causal Analysis for the Impacts of Climate Change on Natural Disaster: with Case Study in Wildfires"*
- *Sarah Löhnert (BS in Information Systems, Summer 2023), "The Causal Relations of Renewable Energy on the Electricity Price: A Case Study based on Solar Energy in Germany"*
- *Tobias Becker (BS in Information Systems, Summer 2023) "Sentiment Analysis on Twitter Based on Machine Learning - A Case Study of Russia's War in Ukraine"*
- *Tim Suchan (BS in Information Systems, Winter 23/24) "Detection of ADHD on Social Media Using BERT-based Time Series Classification"*

Reference

- ***Prof. Jan Jianjun Shi**, Carolyn J. Stewart Chair and Professor*
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- ***Prof. Fugee Tsung**, Chair Professor, Acting Dean*
  Department of Industrial Engineering and Decision Analytics
  The Hong Kong University of Science and Technology, Hong Kong
  Email: season@ust.hk, Phone: +852 2358-7097
- ***Prof. Hao Yan**, Associate Professor*
  School of Computing, Informatics & Decision Systems Engineering
  Arizona State University, U.S.A
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  Department of Geodesy and Geoinformatics
  HafenCity University Hamburg, Germany
  Email: youness.dehbi@hcu-hamburg.de, Phone: +49 40 428 27 - 5328